International Journal of Health Care Finance and Economics, 5, 147–164, 2005
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Is Health Care a Luxury? New Evidence from OECD Data

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Most research using cross-country data find income elasticities equal to or exceeding unity with respect to health expenditure. These conclusions might be confounded due to omitted variables bias and the presence of unobserved country and year specific determinants of per capita health expenditures. I obtain results supporting these hypotheses using recent (1990–98) data from fifteen OECD countries. Specifically, OLS coefficient estimates drop by more than 50% with the use of two-way fixed effects models and the inclusion of various demand and supply based determinants of per capita health expenditures, implying income elasticities of between 0.21 and 0.51. Weighted Least Squares (WLS), Generalized Least Squares (GLS) and Instrumental Variables (IV) estimation yield similar results.

Keywords: health expenditure, income

JEL classification: I18

1. Introduction

Recent research has focused on uncovering significant determinants of health care expenditures. Rising health expenditure to Gross Domestic Product (GDP) ratios across many industrialized countries have imposed stress on government coffers.[1] An idea of the relative impact of different factors affecting trends in government health expenditure would help promote fiscal responsibility by giving policymakers some guidance on appropriate cost containment measures, and consequently result in efficient and innovative methods of financing the system.

Most cross-country studies find per capita income to be the most important determinant of per capita health expenditure. Further, the coefficient estimate of per capita income is equal to or greater than one, leading to the conclusion that health care is a luxury rather than a necessity.[2] Intuitively, one would expect income elasticities with respect to health care to be less than unity, or closer to being a necessity than a luxury. A high-income elasticity is surprising when one considers that most cross-country studies have employed data from countries with public-insurance programs, which guarantee similar standards of access to health care and treatment to all citizens, irrespective of income.

Omitted variables bias could be a possible reason for the large income elasticities obtained in previous studies. Specifically, some studies have employed rather sparse empirical

specifications to identify movements in health expenditure. This is unsurprising, as it is difficult to construct a diverse set of comparable explanatory variables in order to control for demand and/or cost induced shifts of health expenditure. On the other hand, coefficient estimates of per capita income might also be biased upwards because of limited time-series variation available to previous studies. As a result, coefficient estimates of the impact of per capita income will be inflated by an inability to disentangle actual movements in per capita income from year specific shocks that are common across countries. Small sample sizes may also preclude the use of country specific fixed effects and linear trends, thus further confounding empirical estimates of income elasticity due to the impact of unobserved country specific determinants of heath expenditure.

This research contributes to the literature by evaluating empirically the impact of per capita income on trends in health expenditure in fifteen OECD countries between 1990 and 1998 and controlling for a variety of other possible determinants. Recent data availability allows me to lessen the magnitude of omitted variables bias and evaluate the relative importance of demand versus supply factors. Further, the employment of cross-country data over a relatively longer time span, allows me to use two-way fixed effects models, thus separating unobserved country and time invariant shifts of health expenditure from corresponding movements in per capita income. The use of these data also enables me to control for the effects of unobserved time-varying determinants of health expenditures through country specific linear trends.

I find that coefficient estimates of the impact of per capita income decline in magnitude and precision with the inclusion of other observed determinants of per capita health expenditure, country and year fixed effects, and linear trends. Specifically, OLS coefficient estimates fall by more than 50% with these controls. More importantly, my research suggests income elasticities between 0.21 and 0.51, in contrast to findings of equal to or greater than unity obtained by many previous studies. Further, empirical estimates obtained from Weighted Least Squares (WLS), Generalized Least Squares (GLS) and Instrumental Variables (IV) are consistent with this interval. Prior coefficient estimates of the effects of per capita income on health expenditure may therefore, be confounded due to the predominantly cross-sectional nature of most cross-country data sets and the lack of sufficient control variables.

The remainder of the paper is organized as follows. Section 2 elaborates on previous literature. Section 3 gives the empirical specification, and discusses the data. Estimation results are in Section 4. Finally, section 5 concludes with a summary of the main results.

2. Previous Research[3]

Gbesemete and Gerdtham (1992) estimate the impact of per capita income on per capita health expenditure with 1984 data from 30 African countries. They find per capita income to be the most significant determinant of per capita health expenditures. Further, the coefficient estimate of per capita income is statistically significant and roughly equal to one.[4] Gerdtham et al. (1992a) conducted a similar exercise employing 1987 data from 19 OECD countries. Based on coefficient estimates of per capita income, the authors find an income elasticity of -1.33.[5]

The above studies possess some obvious shortcomings. First, the small sample sizes preclude any benefits from asymptotic properties. Second, the reliance on cross-country data from a single year makes it impossible to control for unobserved time-specific shocks that simultaneously affect all countries. Further, by significantly reducing the available degrees of freedom, the small sample sizes also makes it difficult to control for unobserved country specific determinants of health expenditure with the help of fixed effects. Hence, the likelihood of bias in coefficient estimates of per capita income is quite high.

Hitiris and Posnett (1992) contribute to the literature by pooling cross country observations from 20 OECD countries over 28 years, between 1960 and 1987 for a total of 560 observations. They also attempt to control for the effects of unobserved country specific characteristics with country fixed effects. Their results are similar to previous findings, as they also find an income elasticity of around one. However, this result is suspect because of the absence of year fixed effects. The coefficient estimate of per capita income might also be biased upwards by the relative lack of appropriate controls, as the proportion of population over 65 is the only other explanatory variable.

Gerdtham et al. (1992b) estimate the impact of per capita income on per capita health expenditure by employing data from 10 OECD countries for 1974, 1980, and 1987 resulting in a total of 57 observations. So although this study has fewer observations than Hitiris and Posnett (1992), it does have more than earlier studies and employs more covariates.[6] The authors also use year fixed effects. The results of this study indicate an income elasticity exceeding one. However, given the lack of country fixed effects, this estimate may be confounded by unobserved country specific factors.

It is also important to acknowledge the existence of literature, which evaluates the relationship between per capita income and health expenditures by exploiting time-series variation within countries. However, the focus of these studies is on the appropriateness of pooling cross-country data. Employing data from nineteen-sixty to the early nineteen-nineties, some of this research (Hansen and King, 1996; Blomqvist and Carter, 1997; Gerdtham and Lothgren, 2000) find the pooling of OECD panel data to be questionable because of different orders of integration in variables across countries and the presence of unit roots in the per capita health expenditure and income variables. However, using similar data, McCoskey and Selden (1998) reject the null hypothesis of unit roots for both these variables. Of this research, only Blomqvist and Carter (1997) report an income elasticity, which does not seem significantly different from one.[7]

The above discussion illustrates how most previous cross-country studies have focused on estimating the relationship between per capita health care and real income in a base specification with few explanatory variables and limited time-series variation, which precludes the use of either country or year fixed effects or linear trends. As a consequence, the common finding of an income elasticity equal to or exceeding one could be the product of omitted variables bias and/or the impact of unobserved country and time specific determinants of per capita health expenditures. Further, it is difficult to believe that unobserved differences in health policies across countries over time do not significantly impact trends in per capita health expenditures. Good examples are differences in the provision of high-technology

services, such as magnetic resonance imaging and the growth of managed competition in health care.[8]

A recent study by Gerdtham et al. (1998) attempts to remedy these problems by pooling information from 22 countries from 1970 to 1991. The focus of this paper is on evaluating the effects of various institutional arrangements across countries.[9] Of these, only having physicians as gatekeepers is consistently statistically significant and associated with lower health expenditures.[10] Interestingly, the coefficient estimate of per capita remains quite high, ranging from 0.7 to 0.8 across various specifications.

This paper contributes to the literature by using more recent (1990–98) cross-country OECD data to evaluate the relationship between per capita income and health expenditures. The use of these data permits me to control for the effects of a variety of *non-institutional* demand and supply covariates. In particular, this study introduces some new supply or cost based variables. This is consistent with Gerdtham and Jonsson (2000) who have pointed out the need for such research.[11] Although there may be some concerns on pooling data across countries given the obvious differences in data collection and variable definitions, Reinhardt, Hussey and Anderson (2002) suggest that OECD data are quite reliable and more importantly, comparable.[12]

3. Data and Empirical Model

I employ data from the following fifteen OECD countries between 1990 and 1998: Australia, Austria, Canada, Czech Republic, Denmark, Finland, France, Germany, Italy, New Zealand, Poland, Portugal, Sweden, the United Kingdom, and the United States.[13] There are missing observations for Germany in nineteen-ninety. As a result the data consists of 134 observations and is an unbalanced panel of cross-country time-series. The choice of countries is governed by data availability for variables present in the below empirical specification;

$$\begin{aligned}\mathrm{PHE}_{it} &= \beta_0 + \beta_1 PCI_{it} + \beta_2 \mathrm{Demand}_{it} + \beta_3 \mathrm{Supply}_{it} \\ &\quad + \Sigma_t Y_t + \Sigma_i P_i + \Sigma_{it} T_{it} + u_{it} \end{aligned} \tag{1}$$

where PHE_{it} is real per capita health expenditures (public and private); PCI_{it} is real per capita gross domestic product; Demand_{it} represents other demand side variables, which are infant mortality rates per 100,000 population, and the percentage of population above 65 years; Supply_{it} denotes the following variables: average length of patient stay in hospitals and the number of physicians per 1,000 of population. Finally, Y_t are year fixed effects, P_i are country fixed effects, T_i are country specific linear trends and u_{it} is the error term.

The dependent variable in this study is per capita total (public and private) health expenditures (PHE_{it}). The key variable of interest is real per capita income (PCI_{it}), which is represented by real per capita Gross Domestic Product.[14,15] Real per capita income has more or less been on an upward trend over these years. On the surface, Figure 1 suggests that movements in per capita health expenditure appear to be on average, positively correlated with trends in per capita income. However, trends in per capita health expenditure

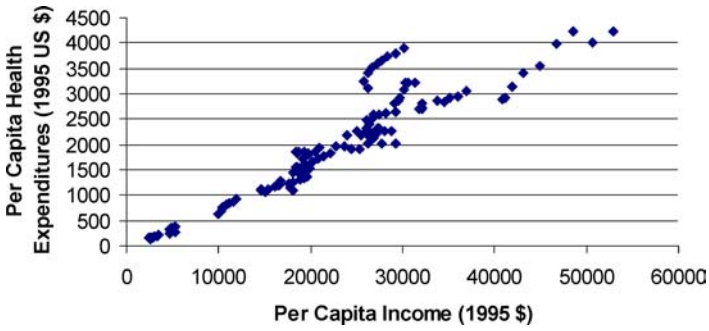

Figure 1. Per Capita Income and Health Expenditure-OECD. The above figure is based on annual averages for fifteen OECD nations between 1990 and 1998. These countries are Australia, Austria, Canada, Czech Republic, Denmark, Finland, France, Germany, Italy, New Zealand, Poland, Portugal, Sweden, United Kingdom, and the United States.

may be driven other factors. A demand side determinant of government health expenditure should be the rate at which people fall sick and/or are injured. A less healthy population will, on average, require more resources and thus result in greater demand for health care and an increase in per capita health expenditures. In order to capture this effect, I employ infant mortality rates (IMR_{it}), which is a commonly utilized measure of health outcomes.[16,17]

Another widely perceived determinant of health expenditure is the ageing population. Age composition in many Western nations has been changing as the baby boomers begin to age, and life expectancy rates increase. One view is that an ageing population imposes significant public health care expenses, as the elderly are usually cared for the remainder of their lives. Therefore, an increase in the percentage of elderly people results in a more than proportionate increase in health care costs. In contrast, some argue that health care costs increase with proximity to death, and costs are otherwise constant throughout the senior years.[18] In both cases, an ageing population is expected to impose a significant burden on government health expenditure. On the other hand, the percentage of elderly people may be negatively correlated with government health expenditure as voter participation declines with age, rendering the interests of elderly people less important to administrators.[19] The relationship between government health expenditures and the percentage of elderly people is, therefore, ambiguous. $P65_{it}$ is the proportion of population aged sixty-five and over, and is constructed to evaluate this relationship.

Trends in per capita health expenditures should also be influenced by supply side or cost based variables. There is a voluminous literature, which cites increases in average lengths of stay ($ALOS_{it}$) of hospital patients as an important cost factor determining the provision of health services. Most of these studies are focused on evaluating these costs against possible improved health outcomes for patients through longer hospital stay.[20]

Another important supply side determinant of per capita health expenditures is physician costs, which can be represented by the number of physicians per 1,000 of population (NOP_{it}), and should capture trends in patient care access. There exists evidence on restrictions on physician supply as a component of reducing health services costs.[21] Of course,

Table 1. Summary statistics.

Name	*N*	Mean	St. dev	Minimum	Maximum
Real per capita health expenditure	134	1892.3	1051	138.03	4237.92
Real per capita GDP (PCI_{it})	134	21842.	10826	2414.0	52974
Infant mortality rates (IMR_{it})	134	7.0784	2.8840	3.5000	19.300
Percentage of population over age 65 ($P65_{it}$)	134	14.020	2.0294	10.055	17.794
Average length of stay ($ALOS_{it}$)	134	11.145	2.8895	6.5000	18.800
Number of physicians per 1000 of population (NOP_{it})	134	2.7769	0.82916	1.4000	5.8000
Per capita alcohol consumption (ALC_{it})	118	10.213	2.4871	5.8000	16.600
Percentage of smokers	118	26.642	4.5451	19.100	44.500

compensation methods play an important role in determining the effects of physician supply on health expenditures. For example, an increase in salaried physicians should result in higher per capita health expenditures. On the other hand, an increase in physicians might not impact health costs under a fee-for-service scheme if there is a migration of patients from existing to new practitioners. However, per capita health expenditures may rise if existing and established practitioners respond by encouraging more frequent visits from patients, in an effort to maintain unchanged incomes.

Table 2 summarizes cross-province and time-series variation in OECD data. Part A captures cross-province variation in the dependent variable and all covariates by calculating the percentage difference between the maximum and minimum values of a variable in beginning and ending sample years.[22] On the other hand, time-series variation is captured in part B, which calculates the difference in sample mean values for each variable between 1991 and 1998.[23]

The figures in Table 2 suggest greater cross-country than time-series variation. This is because cross-country variation between maximum and minimum values across countries within sample end points is larger for almost each variable than time-series variation. For example, cross-country variation in per capita income (PCI_{it}) was 94.08% and 93.59% in 1990 and 1998, respectively (column 2). In comparison, average real per capita income (PCI_{it}) increased by 15.51% between these years. These figures suggest significant cross-country differences accompanied by much more moderate time series movements. Hence, to further ensure the robustness of our empirical results, I use country fixed effects to minimize the potentially confounding effects of unobserved country specific attributes. Country linear trends are used to capture the impact of unobserved nation specific determinants of per capita health expenditures that are time varying. Finally, year fixed effects are employed to account for factors that may similarly impact all countries in a specific year.

4. Empirical Results

This section sets an empirical baseline for evaluating the impact of per capita income on trends in government health expenditure. Table 3 contains OLS estimates of equation (1)

Table 2. Cross country and time-series variation—OECD data.

	Per capita income (1995 US $) (1)	Per capita health exp (1995 US $) (2)	Percent of pop over 65 (3)	Average length of stay (4)	Infant mortality rates (5)	Number of physicians per 1000 of Pop. (6)
			A. cross-country variation			
Difference across countries—1998						
Max	52974	4237.9	17.55	15.9	19.3	5.8
Min	3396	217.34	11.60	6.6	3.5	1.7
% Difference	**−93.59**	**−94.87**	**−33.90**	**−58.49**	**−63.16**	**−70.69**
Difference across countries–1990						
Max	41179	3113.0	17.794	18.2	19.3	4.7
Min	2604.3	138.03	10.055	8.2	5.6	1.4
% Difference	**−94.08**	**−95.09**	**−42.25**	**−57.45**	**−67.58**	**−71.43**
			B. Time-series variation			
Mean values—1991	20721.44	1767.01	13.67	12.94	8.44	2.63
	23935.73	2175.14	14.46	8.91	5.03	2.95
Mean values—1998						
% Change	**15.51**	**23.097**	**5.82**	**−31.12**	**−40.36**	**12.15**

from OECD data, using the natural logarithm of real per capita provincial health expenditure (PHE_{it}) as the dependent variable.[24] The natural logarithm of per capita income (PCI_{it})) is the sole explanatory variable in column 1; columns 2 and 3 add country and year fixed effects respectively; column 4 includes demand side variables (percentage of population over 65 ($P65_{it}$) and infant mortality rates (IMR_{it})); column 5 adds supply-side factors (average length of patient stay ($ALOS_{it}$), and the number of physicians per 1,000 of population (NOP_{it})); and finally, column 6 includes country-specific linear trends. The null hypothesis of no first or second-order autocorrelation could not be rejected in all regressions.[25] In contrast, the null hypothesis of homoscedasticity is rejected by a variety of tests.[26] Hence, I focus on correcting standard errors for heteroskedasticity. Given its unknown nature, I rely on the procedure recommended by White (1980).

The first noteworthy result is the statistically significant (at 1%) and greater than unity coefficient estimate of per capita income (PCI_{it}) from column 1, implying that health care is a luxury. Further, per capita income (PCI_{it}) explains about 97% (adjusted R-square) of the variation in per capita health expenditures (PHE_{it}). As discussed earlier, this is consistent with previous studies that have used OECD data and relied on single year data points across countries to identify the relationship between per capita income and government health expenditure. Column 2 attempts to control for the confounding effects of unobserved determinants of per capita health expenditures through the use of country fixed effects, which are jointly significant at 1%. The coefficient estimate of per capita income (PCI_{it}) remains relatively unchanged and statistically significant at 1%. The addition of year fixed effects in

Table 3. OLS regression results—OECD countries[a].

Explanatory variables	Per capita income (1)	Country fixed effects (2)	Year fixed effects (3)	Demand variables (4)	Supply side variables (5)	Country specific linear trends (6)
Real per capita GDP (PCI_{it})	**1.1593** **(0.01586)*****[43]	**1.0610** **(0.09899)*****	**0.81886** **(0.21355)*****	**0.80177** **(0.21670)*****	**0.53206** **(0.20806)*****	**0.50976** **(0.20099)*****
Percentage of population over age 65 ($P65_{it}$)				0.04218 (0.21225)	−0.23700 (0.2229)	−0.83374 (0.56681)
Infant mortality rates (IMR_{it})				−0.02566 (0.0643)	−0.07335 (0.06411)	−0.04389 (0.02974)
Average length of stay ($ALOS_{it}$)					**0.24983** **(0.05868)*****	0.05761 (0.04923)
Number of physicians per 1000 of population (NOP_{it})					0.23101 (0.16051)	0.3625 (0.24852)
Number of observations	134	134	134	134	134	
Year fixed effects	NO	NO	YES/ $p-$ value = 0.03775	YES/ $p-$ value = 0.08418	YES/ $p-$ value = 0.02905	YES/ $p-$ value = 0.03299
Country fixed effects	NO	YES/ $p-$ value = 0.00000	YES/ $p-$ value = 0.00000	YES/ $p-$ value = 0.00000	YES/ $p-$ value = 0.00000	YES/ $p-$value = 0.00000
Country linear trends	NO	NO	NO	NO	NO	YES/ $p-$ value = 0.00000
Adjusted *R*-square	0.9669	0.9938	0.9944	0.9943	0.9950	0.9977

***Significant at 1%, **Significant at 5%, *Significant at 10%.

[a]Standard errors are found in parentheses below estimated coefficients and are White-corrected for hetteroskedasticity. Empirical results were obtained by regressing the above variables on per capita health expenditures for fifteen OECD nations between 1990 and 1998. These countries are Australia, Austria, Canada, Czech Republic, Denmark, Finland, France, Germany, Italy, New Zealand, Poland, Portugal, Sweden, United Kingdom, and the United States. There were missing observations for Germany in 1990, leaving 134 observations.

column 3, which are intended to account for the effects of time-specific shocks experienced by all countries, further reduces the coefficient estimate of per capita income by roughly 23% to 0.81886. These year-fixed effects are jointly significant at 5%. The magnitude and precision of the coefficient estimate of per capita income (PCI_{it}) drops slightly to 0.80177 with the inclusion of other demand side variables (percentage of population over 65 ($P65_{it}$) and infant mortality rates (IMR_{it})) in column 4. The coefficient estimates of both these variables are statistically imprecise. On the other hand, country and year fixed effects are statistically significant.

Column 5 focuses on the consequences of including supply side variables (average length of stay ($ALOS_{it}$) and number of physicians (NOP_{it})). Per capita income (PCI_{it}) remains statistically significant at 1% with the addition of supply side variables, but more interestingly, falls by roughly 50% to 0.53206. The coefficient estimates of the percentage of elderly people ($P65_{it}$) and infant mortality rates (IMR_{it}) are negative and statistically insignificant. The negative coefficients estimates on both these variables could be a result of

the effects of other unobserved factors, which are negatively correlated with trends in per capita health expenditures. In terms of supply side variables, the average length of hospital stay ($ALOS_{it}$) is statistically insignificant and positively correlated with trends in government health expenditure. In contrast, the coefficient estimate of the number of per capita physicians (NOP_{it}) is statistically significant. Country and year fixed effects remain jointly significant. Finally, column 6 evaluates the effects of including country specific linear trends. The coefficient estimate of per capita income (PCI_{it}) drops slightly to 0.50976 but is still statistically significant at 1%. On the other hand, the number of physicians (NOP_{it}) is no longer statistically significant. Finally, country and year fixed effects and country specific trends are all jointly significant.

In summary, results from Table 3 demonstrate the merits of including new supply side variables ($ALOS_{it}$ and NOB_{it}) in mitigating omitted variables bias. Although the coefficient estimates of these variables eventually become statistically insignificant, their inclusion considerably reduces the magnitude of the coefficient estimate of per capita income (PCI_{it}). Further, employing two-way fixed effects models enables us to disentangle the effects of per capita income from other unobserved determinants of health expenditure that are time invariant within countries and simultaneously impact different countries within time. Finally, country specific linear trends are important to account for unobserved factors, which are time-varying within countries.

The joint significance of country and year fixed effects and country linear trends is unsurprising given the wide array of unobserved factors that could plausibly impact trends in per capita health expenditure. For example, although most OECD countries have public health insurance, there are differences in the magnitude of coverage.[27] Also, there exist significant differences in the use of medical technology across countries. Similarly, there are often considerable cross-country differences in caps imposed on billings submitted by physicians operating within fee-for-service systems. The addition of these covariates results in approximately a 56% decline in the magnitude of the coefficient estimate of per capita income (PCI_{it}), from 1.16 to 0.5076, implying that health care is not a luxury.[28]

The preceding discussion suggests omitted variables bias, in particular arising from the exclusion of supply-side variables as the main reason for biased coefficient estimates of per capita income (PCI_{it}). However, the benefits of possessing cross-country data over a longer time dimension are clearly brought out in columns 1 and 2 of Table 4. Specifically, I conduct a counter-factual exercise, by replicating the analysis performed by Gerdtham et al. (1992b), who estimate the impact of per capita income (PCI_{it}) by employing data from 10 OECD countries for 1974, 1980, and 1987. Column 1 evaluates the effect of per capita income (PCI_{it}) on per capita health expenditures (PHE_{it}) by following their basic format, which is achieved by employing other demand and supply based determinants and year fixed effects but not country fixed effects nor linear trends. The data is arbitrarily drawn from three years: 1990, 1993, and 1996. Column 2 consists of a similar exercise with data from 1991, 1994, and 1997.

The results across these columns are very similar. While per capita income is statistically significant at 1%, the coefficient estimates of 1.2598 (column 1) and 1.2411 (column 2) clearly imply that health care is a luxury and are consistent with the 1.26 value obtained by Gerdtham et al. (1992b). The magnitude and precision of the coefficient estimate of

Table 4. OLS Regression results–sensitivity checks.[a]

Explanatory variables	Data from 1990, 1993, 1996 **(1)**	Data from 1991, 1994, 1997 (2)	Levels model (3)	Semi-log model (4)
Real per capita GDP (PCI_{it}) [Implied Elasticity]	**1.2598** **(0.0598***)**[45]	**1.2411** **(0.05799)*****	**0.024711** **(0.01367)**** [0.2852]	**603.55** **(228.02)***** [0.3189]
Percentage of population over age 65 ($P65_{it}$)	**−0.62975** **(0.16107)*****	**−0.65688** **(0.17224)*****	−68.115 (83.892)	−83.613 (92.194)
Infant mortality rates (IMR_{it})	0.026747 (0.11722)	1.2411 (0.05799)	3.7333 (3.0090)	43.236 (44.715)
Average length of stay ($ALOS_{it}$)	**0.18172** **(0.09312)****	**0.19209** **(0.08805)****	**17.945** **(7.2818)****	**204.63** **(94.072)****
Number of physicians per 1000 of population (NOP_{it})	**0.24779** **(0.06092)*****	**0.23030** **(0.06233)*****	121.83 (100.65)	192.54 (247.98)
Number of observations	44	45	134	134
Year fixed effects	YES	YES	YES	YES
Country fixed effects	NO	NO	YES	YES
Country linear trends	NO	NO	YES	YES
Adjusted *R*-square	0.9753	0.9742	0.9969	0.9961

***Significant at 1%, **Significant at 5%, *Significant at 10%.

[a]Standard errors are found in parentheses below estimated coefficients and are White-corrected for hetteroskedasticity. Columns 1 and 2 consist of empirical results obtained by regressing the above variables on per capita health expenditures for fifteen OECD nations for the years 1990, 1993, and 1996; and for the years 1991, 1994, and 1997, respectively. Columns 3 and 4 consist of results based on data from fifteen OECD countries between 1990 and 1998. These countries are Australia, Austria, Canada, Czech Republic, Denmark, Finland, France, Germany, Italy, New Zealand, Poland, Portugal, Sweden, United Kingdom, and the United States. There were missing observations for Germany in 1990, leaving 44 observations for column 1 and 134 observations for columns 3 and 4.

the average length of hospital stay ($ALOS_{it}$) is quite similar to the corresponding estimate from column 5 of the previous table. The number of physicians (NOP_{it}) is also statistically significant and is quite similar to the corresponding value in column 5, Table 3. Finally, the counter-intuitive negative sign of the percentage of elderly ($P65_{it}$) indicates the presence of unobserved factors that are negatively correlated with per capita health expenditures.

Hence, coefficient estimates of per capita income (PCI_{it}) and most other explanatory variables seem to be biased upwards in magnitude and precision due to the absence of any controls for unobserved determinants of per capita health expenditures, which are time-invariant and time-varying within countries. In contrast, as demonstrated by the empirical estimates contained in Table 3, the use of data over a relatively longer time span permits the clearer identification of year specific shocks as well as the use of country fixed effects and country linear trends due to increased sample size.

The remaining columns in Table 4 consist of other sensitivity checks. Specifically, columns 3 and 4 consist of empirical estimates from levels and semi-log versions of equation (1). As mentioned earlier, results from likelihood-ratio tests support the use of a log-log model. Nonetheless, similar results from other types of specifications will strengthen the

Table 5. WLS, GLS, and IV results.[a]

Explanatory Variables	Weighted least Squares (WLS) (1)	Generalized least Squares (GLS) (2)	instrumental Variables (IV) (3)
Real per capita GDP (PCI_{it})	**0.37678** **(0.21681)***	**0.32723** **(0.13664)****	0.17603 (0.51341)
Percentage of Population over age 65 ($P65_{it}$)	0.38795 (0.63937)	−0.028003 (0.59089)	−0.87589 (0.55324)
Infant Mortality rates (IMR_{it})	0.021558 (0.07498)	0.035510 (0.025199)	−0.16257 (0.24784)
Average length of stay ($ALOS_{it}$)	0.096416 (0.069186)	**0.091441** **(0.040226)****	**0.92436** **(0.51890)***
Number of physicians per 1000 of population (NOP_{it})	0.10585 (0.21543)	**0.28061** **(0.14638)***	0.41462 (0.31030)
Number of observations	134	134	118
Year fixed effects	YES	YES	YES
Country fixed effects	YES	YES	YES
Country linear trends	YES	YES	NO
Adjusted R-square	0.9988	0.9984	0.9905

***Significant at 1%, **Significant at 5%, *Significant at 10%.

[a]The data consists of Australia, Austria, Canada, Czech Republic, Denmark, Finland, France, Germany, Italy, New Zealand, Poland, Portugal, Sweden, United Kingdom, and the United States between 1990 and 1998 for columns 1 and 2. There were missing observations for Germany in 1990, leaving 134 observations. Column 3 excludes most observations from Poland and Portugal. Rows consisting of indicators for country and year fixed effects and country linear trends also consist of p-values of F-tests of joint significance.

finding of health care not being a luxury good. The interesting result from these specifications is the fact that implied income elasticities are much lower than the value of 0.5076 obtained from the log-log model. Specifically, the implied elasticity from the levels model is 0.2852, while the corresponding figure from the semi-log model is 0.3189. It is important to note that the average length of stay ($ALOS_{it}$) is also statistically significant across both specifications. In summary, the different sensitivity tests embodied across the columns in Table 4 give strikingly similar results. Specifically, while per capita income (PCI_{it}) is significantly associated with trends in per capita health expenditures, its coefficient estimate is lower than one, and even lower than the benchmark estimate of 0.5076. Hence, results from alternative specifications offer further evidence against health care being a luxury.

Table 5 consists of sensitivity analyses pertaining to different estimation procedures. As mentioned above, standard errors for all coefficient estimates have been White corrected (1980) for heteroskedasticity. On the other hand, given the significant variation in population across countries, a plausible alternative is the use of Weighted Least Squares (WLS), with observations being weighted according to country population. The results of this exercise are contained in column 1 of Table 5. In terms of statistical precision, coefficient estimates are quite similar to the log-log model (column 6, Table 3) as only per capita income (PCI_{it}) is statistically significant. However, the coefficient estimate of PCI_{it}, at 0.3768 is much lower than the corresponding OLS coefficient estimate of 0.50976.

As a further sensitivity test, Generalized Least Squares (GLS) was also employed in order to correct coefficient estimates for heteroskedasticity. These estimates are contained in column 2 of Table 5.[29] The coefficient estimate of per capita income is quite similar to the corresponding WLS estimate and is statistically significant. Average length of stay ($ALOS_{it}$) and the number of physicians (NOP_{it}) are also positive and statistically significant at 5 and 10%, respectively. The important point is that correcting for heteroskedasticity through different methodologies does not change earlier findings. Specifically, that while per capita income is significantly correlated with trends in per capita health expenditures, its coefficient estimate is significantly lower than one.

The empirical results obtained so far suggest that health care is not a luxury good. However, placing emphasis on results from OLS coefficient estimates could be misguided strategy because of possible simultaneity bias. In other words, some of the exogenous variables specified on the right hand side of equation (1) might be jointly determined with per capita health expenditures, which would result in a correlation with the error term and inconsistent estimates. Equally important, OLS coefficient estimates of these variables would be biased. Consequently, OLS coefficient estimates of the impact of per capita income (PCI_{it}) might also be biased, depending on the correlation between per capita income (PCI_{it}) and these other variables.

For example, an increase in per capita health expenditures might result in significantly lower infant mortality rates (IMR_{it}) because of the availability of better resources. Hence, OLS coefficient estimates of the impact of infant mortality rates (IMR_{it}) will be biased downwards. Coefficient estimates of per capita income (PCI_{it}) will then be biased upwards if per capita income is negatively correlated with infant mortality rates. Similarly, the average length of hospital stay ($ALOS_{it}$) may also be simultaneously determined along with health expenditures. Increases in health expenditure could lead to new hospitals and consequently more facilities, thus permitting increased tenure of hospital stay. The resulting bias in OLS coefficient estimates of the average length of stay ($ALOS_{it}$) could then impact the magnitude of corresponding OLS estimates of the effects of per capita income (PCI_{it}) on health expenditures, subject to the relationship between it and the average length of stay.

On the other hand, simultaneity in OLS estimates of the percentage of elderly people ($P65_{it}$) and the number of per capita physicians (NOP_{it}) is less likely. Studies have found a statistically significant link between health expenditure and life expectancy.[30] However, the present pool of elderly people is likely due to improvements in lifestyle, nutrition, and medical technology, which have occurred over the past several years. The probability of a contemporaneous correlation between the two is quite low. A similar argument holds for the number of physicians (NOP_{it}). An increase in health expenditure, whether it takes, for example, the form of higher physician reimbursement rates or more medical schools should not have an immediate impact on the number of physicians given the considerable time lag between their education and training, and actual practice.

On the other hand, it is possible that shrinking health expenditures could immediately impact physician supply through restrictions on the entry of foreign trained physicians. However, there are only three countries (Australia, Canada, and the United States) in the sample with a significant number of foreign trained physicians.[31] As a quick and crude check on the possibility of simultaneity bias, I ran OLS regressions without these countries

(similar to the full specification contained in column 6, Table 3) and compared the results to regressions with the full sample. This counterfactual exercise is predicated on the intuition that simultaneity bias should result in significantly different coefficient estimates of the number of physicians when countries with a relatively high number of foreign physicians are excluded. However, empirical estimates of the number of physicians as well as other variables are in fact similar across both samples, casting doubt on the possibility of simultaneity bias.[32]

The above discussion suggests the need for multiple instruments, which are correlated with potentially simultaneously determined variables but not with the right hand side error term. Alcohol consumption (in litres) per capita of population and the (national) percentage of smokers are used to explain trends in both infant mortality rates (IMR_{it}) and the average length of hospital stay (ALOS_{it}) independent of contemporaneous trends in health expenditure.[33] Both alcohol and cigarette consumption (or variants of them) have been found to significantly impact mortality rates and the duration of hospital stay by many studies.[34]

The theory behind employing alcohol consumption and the percentage of smokers as instruments is derived from the well established fact that lifestyle choice is an important component with respect to the production of health, and hence population health.[35] In this specific case, alcohol consumption can lead to liver disease, while tobacco use is associated with lung disease and cancer, which obviously impacts hospital stay. Within a shorter time span, alcohol consumption is strongly correlated with an increase in being involved in a motor vehicle accident, which also affects the length of hospital stay. On the other hand, alcohol consumption is linked to infant mortality through prenatal use by pregnant women.[36] Tobacco use by pregnant women is also associated with increased infant mortality. In summary, alcohol consumption and the percentage of smokers are plausible instruments because they impact per capita health expenditures indirectly through higher infant mortality and increased stay at hospitals, and hence should not be correlated with the right hand side error term in equation (1).[37]

The appropriateness of the above instruments is tested using F- tests of the null hypothesis that the true relationship between the instruments and potential endogenous variables is zero.[38] The null hypothesis is rejected for all covariates.[39] Further, the Hausman test rejects at the 5% level, the null hypothesis that infant mortality rates (IMR_{it}) and the average length of hospital stay (ALOS_{it}) are exogenous.[40] IV estimation results are given in column 3 of Table 5. The coefficient estimate of per capita income (PCI_{it}) is statistically insignificant, but at 0.17603 is significantly lower than one. With the exception of the average length of stay (ALOS_{it}), all other variables are statistically insignificant. The statistical significance of the average length of stay (ALOS_{it}) is interesting given similar results across other (levels, semi-log, GLS) specifications.

It must be emphasized that the above analyses cannot be viewed as a fully satisfactory solution to the potentially confounding effects of simultaneity bias. Employing more instruments in order to further evaluate the sensitivity of empirical estimates would definitely be a worthwhile exercise. Further, the availability of more instruments would possibly enable a more rigorous modelling of the complex relationships existing among the variables employed in this research through a comprehensive simultaneous equations model. However,

the use of cross-country data makes this very difficult. In many instances, similar data are simply unavailable. If available, it is rare that the data are comparable in terms of collection protocols and statistical methodologies. Given these limitations, the IV estimates obtained in this research should be viewed with caution. But given the objective of the paper, it is reassuring that employing plausible instruments results in coefficient estimates of per capita income that are significantly less than one and consistent with corresponding OLS results.

5. Conclusion

Most estimates from cross-country studies have found income elasticities with respect to health care to be equal to or greater than unity. In contrast, this research suggests a range of 0.21 and 0.51, implying that previous findings might be biased due to omitted variables and unobserved country and year specific determinants of per capita health expenditures. This is because previous studies have mostly used cross-country data with limited time-series variation and a lack of adequate controls to proxy the effects of demand and supply or cost based determinants of government health expenditure. Further, WLS, GLS and IV estimates are consistent with corresponding OLS results.

Acknowledgment

I would like to thank Mark Pauly, two anonymous referees, and Kevin Milligan for some very useful comments and Anthony Clemente, Mark Bergman and Aristides Gianneskis for excellent research assistance. The usual disclaimer applies.

Notes

1. Between 1960 and 1998, health expenditures as a proportion of total Gross Domestic Product (GDP) increased from 4.3 to 8% for Australia, 5.4 to 9.3% for Canada, 4.1 to 9.4% for France, 4.5 to 7.9% for Sweden, 3.9 to 6.8% for the United Kingdom, and 5.1 to 12.9% for the United States (OECD Health Database 2001).
2. Newhouse (1977) employed data on per capita medical expenditures for 13 countries in 1970, Leu (1986) used information on 19 OECD countries from 1974, Brown (1987) used a sample of 20 OECD countries from 1978, Gerdtham et al. (1992a) based their results on a data set of 19 OECD countries in 1987, Gbesemete and Gerdtham (1992) utilized a 1984 sample of 30 African countries, Hitiris and Posnett (1992) pooled observations from 20 OECD countries between 1960 to 1987, while Gerdtham et al. (1992b) employed data from 10 OECD countries for 1974, 1980, and 1987.
3. This discussion focuses on more recent research. Earlier studies are cited in footnote 3.
4. The percentage of births attended by health staff and per capita foreign aid are also significantly and positively correlated with trends in per capita health expenditure.
5. However, coefficient estimates of the impact of urbanization, share of public health expenditure, share of inpatient care expenditure, and a fee-for-service fixed effect are also statistically significant.
6. The authors construct controls for urbanization, proportion of elderly, female labour participation rate, number of per capita physicians, percentage of inpatient health care expenditure, and fixed effects for countries with fee for service remuneration.
7. Specifically, they find an income elasticity of 0.967.
8. For a detailed account of these differences, please see OECD (1994).

9. Institutional features that appear as explanatory variables include: public reimbursement and public integrated systems, budget ceilings for the hospital and ambulatory sectors, physicians as gatekeepers, the prevalence of fee-for-service systems, capitation, and whether physicians are reimbursed through fixed wages. Covariates also exist for the proportion of elderly and children four years and under, per capita alcohol and tobacco consumption, female labour participation rate, unemployment rates, number of per capita physicians, and the percentage of inpatient health care expenditure. Finally, country and year specific fixed effects are also employed.
10. In terms of other robust results, in-patient care expenditures (as a proportion of total health expenditure) is significantly correlated with higher costs.
11. "Empirical studies in recent years have been remarkably unwilling to test "new" variables as regressors in their models." (p. 48)
12. In their words: "By the standards of statistical reporting routinely accepted by decisionmakers in business, finance, and government for other areas of economic activity, the data reported by the OECD appears to be quite robust", (p. 179).
13. This information was extracted from the OECD Health Statistics Database (http://www.oecd.org/oecd/pages/home/displaygeneral/0,3380,EN-statistics-194-5-no-no-no-194,FF.html). As elaborated in the literature review, data on per capita health expenditures and income are available over a longer time period (from 1960 onwards) and for more countries. However, information on other variables employed in this study, is only more recent.
14. OECD data is in constant 1995 US $.
15. Empirical constructs of variables are represented by acronyms.
16. Theory does not clearly state which variables would be appropriate to include as right hand side covariates in determining trends in per capita health expenditures. Consequently the above empirical specification is primarily based on prior findings from the health literature and is only constrained by data availability.
17. Cremieux, Ouellette and Pilon (1999) evaluates the impact of various socio-economic variables on infant mortality rates across Canadian provinces between 1978 and 1992.
18. For a detailed discussion of these conflicting views, please refer to Robson (2001). Zweifel, Felder and Meiers (1999) provide some empirical evidence on the age neutrality of health care expenditure using Swiss data.
19. This has been suggested by Gouveia (1996).
20. Hayes et al. (2000), Collins et al. (2000), Yao and Lu-Yao (1999), Deaton et al. (1998), Cupuro and Alperovich (1997), and Thomas, Guire and Horvat (1997), and Stern, Singer and Weissmen (1995).
21. In order to reduce the number of practising physicians and to ensure an appropriate physician/population ratio, a national strategy in Canada was announced in 1992 which focussed on reducing medical school entry class size (by 10%); reducing national postgraduate medical training positions by 10%; and reducing recruitment of visa trainee graduates of foreign medical schools into Canada for postgraduate medical training. From 1993 to 1996 the number of physicians decreased relative to the population (Davis, 1999).
22. For example, column 2 shows that the highest per capita income a country enjoyed in 1991 was $41,179, while the lowest was $2604.30. The difference between these two values is 94.08%.
23. Column 2 shows that average per capita income in 1991 across all countries was $20,721.44, while the corresponding value for 1998 was $23,935.73. Hence, real per capita income increased by 15.51% over the sample period. 1991 is chosen as a lower bound due to some missing observations with respect to Germany in 1990.
24. With the exception of binary indicators, all variables are in natural logarithms. Using a log-log specification permits coefficient estimates to be interpreted as elasticities and facilitates the comparison of our results to prior studies. Further, Likelihood Ratio tests based on Box-Cox regressions reject the use of a levels specification.
25. I used Baltagi's modified h-test for panel data. Please see Baltagi (1995) for further details.
26. For example, Breusch-Pagan's and Gjesler's tests. Please see Greene (2001) for further details.
27. Italy, Belgium, Denmark, Finland, Greece, Iceland, Luxembourg, Norway, and Spain have virtually no exclusions, while dental and optical care and pharmaceuticals are largely covered by private insurance in Canada and the United Kingdom. However, there is also cost sharing of health expenditure in countries where there are no restrictions on types of services covered, which is not the case for Canada and the United Kingdom.
28. It is important to emphasize that this relation between cannot be viewed as a causal one; but merely as a strong and consistent association after controlling for other possible determinants of health expenditures.

29. The GLS estimation was based on the method suggested by Parks (1967).
30. Cremieux, Ouellette and Pilon (1999).
31. Bourassa Forcier, Simoens and Giuffrida (2004) suggest that foreign trained practitioners are equal to or more than 20% of the total number of physicians in Australia, Canada, and the United States, and are significantly lower in number for other countries in the sample, ranging from 2 and 11%.
32. The coefficient estimate of the number of physicians with respect to per capita health expenditures from the log-log specification in Table 3, column 6 is 0.3625 with a standard error of 0.24852. Excluding Australia, Canada, and the United States results in a corresponding OLS coefficient estimate of 0.41311 with a standard error of 0.35098. More importantly, the OLS coefficient estimate of per capita income from the reduced sample is 0.52599, which is very similar to the corresponding estimate of 0.5076 from the full specification (Table 3, column 6).
33. As with the earlier data, data on per capita alcohol consumption and the percentage of smokers were obtained from the OECD (www.oecd.org) website.
34. See Cremieux, Ouellette and Pilon (1999) for a summary.
35. Please see Folland, Goodman, Stanos (2004) for a detailed discussion.
36. Faden, Graubard and Dufour (1997).
37. A valid point is that lagged smoking rates and alcohol consumption are also plausible instruments since adverse health outcomes from their consumption often take some time to manifest themselves. Unfortunately due to data limitations we are forced to restrict our instrument set to contemporaneous cigarette and alcohol consumption. Further, this is a strategy, which has been adapted by other research (Cremieux, Ouellette and Pilon 1999; and Cochrane, St. Leger and Moore 1978).
38. Most observations from Poland and Portugal had to be dropped because of unavailable data, reducing the sample size to 118 observations. As a result, country specific trends could not be employed because of consequent collinearity. However, this does not seem to be too important as the OLS coefficient estimate of per capita income from this reduced sample is 0.502, which is almost identical to the corresponding OLS coefficient estimate of 0.51 obtained from the full sample with country specific trends.
39. Specifically, the p-values of the joint test of significance with respect to infant mortality (INFM_{it}) and the average length of stay (ALOS_{it}) are 0.00290 and 0.05583, respectively. Detailed coefficient estimates are available on request.
40. We use the following form of the Hausman test. Suppose $Y = X\beta + \varepsilon$ and Z is a set of instruments for X. With some algebraic manipulation, it can be shown that $\beta^{IV} - \beta^{OLS} = (Z'X)^{-1}Z\prime\varepsilon^{OLS}$, which will be zero if Z and ε^{OLS} are uncorrelated. This can be tested by running the regression $Y = X\beta + Z \quad \theta + \varepsilon$ and testing $\theta = 0$ with an F test The test statistic is distributed with r and $n - k$ degrees of freedom, where r = number of variables in Z, n = number of observations, and k = number of explanatory variables. Please see Kennedy (1997) for further details.

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